Provable Group Fairness via Class-Group Prototype Alignment in Federated Learning

Anonymous Author(s)
Anonymous Institution

Abstract

Group fairness in federated learning (FL) faces three coupled challenges: sensitive group statistics are exposed during communication, representation differences between same-label different-group samples are left unaddressed, and local fairness improvements do not propagate to the global model under statistical heterogeneity. We formulate three falsifiable conjectures and support each with theory: (C1) the representation gap Δ_{rep} between same-label different-group features controls the equalized-odds (EO) *upper* bound; (C2) the prototype-level confidence difference δ_{conf} between group prototypes, as read by the shared classifier, controls the EO *lower* bound; (C3) local fairness and accuracy propagate to the global model only when encoder and classifier are anchored to globally consistent prototype coordinates, with heterogeneity entering the global bound as a mean-prototype-deviation term. Each conjecture yields a measurable prediction (P1–P3), verified empirically. Based on these results we design **PDFFed** (Prototype-Driven Fair Federated Learning), which uses *class-group prototypes* as a unified knowledge carrier spanning client training, communication, and aggregation. PDFFed introduces three mechanisms that realize the conjectures: prototype communication replacing group statistics; feature-space alignment L_{PA} and prototype-level confidence calibration L_{conf} that jointly bound EO from above and below while preserving accuracy; and bidirectional prototype–classifier alignment plus server-side post-training that propagate local gains to the global model. Across image, text, and tabular tasks (7 datasets, 4 heterogeneity levels, 15+ baselines) we verify predictions P1–P3: Δ_{rep} and δ_{conf} decrease as predicted, the global accuracy–fairness Pareto front is improved, and the per-round communication cost is competitive.

1 Introduction

1.1 Motivation: Three Coupled Challenges

- **Perception.** Existing methods upload group statistics (group sizes, proportions, local fairness metrics) to the server, exposing sensitive information at the communication level and suffering from noisy estimates on small sub-groups under statistical heterogeneity.
- **Adjustment.** Existing methods impose fairness constraints in the output space or reweight aggregation; they do not address the representation gap between same-label different-group samples, which is the root cause of group-unfair predictions.
- **Transmission.** Existing methods focus on local optimization; under statistical heterogeneity, local fairness gains do not transfer to global guarantees.

1.2 Three Conjectures

We start from three falsifiable conjectures, stated informally here and formalized in Sec. 4.

1. **Conjecture C1 (perception).** The representation gap Δ_{rep} between same-label different-group features controls the EO *upper* bound: closing Δ_{rep} in the encoder provably closes EO, without touching the classifier. *Measurable prediction P1:* Δ_{rep} decreases under training and stabilizes at a small positive residual, and EO follows it downward (Thm. 6, Thm. 7; verified in Sec. 6).

2. **Conjecture C2 (adjustment).** When the shared classifier systematically responds with high confidence to one group's prototypes and low confidence to the other's, EO is forced to be large; the prototype-level confidence difference δ_{conf} is a faithful, stable surrogate for this lower-bound mechanism. *Measurable prediction P2:* minimizing δ_{conf} lowers DEO / $\text{EO}_{\max}$, and the low/high-confidence scenario is observable in the confidence distributions (Thm. 10; verified in Sec. 6).

3. **Conjecture C3 (transmission).** Local fairness and accuracy improvements propagate to the global model only if encoder and classifier are anchored to globally consistent prototype coordinates; heterogeneity enters the global fairness bound as a mean-prototype-deviation term. *Measurable prediction P3:* under high heterogeneity, PDFFed keeps global accuracy while suppressing global EO, and the global fairness degradation grows at most linearly with the prototype deviation $\bar{\Gamma}$ (Lem. 13, Thm. 14; verified in Sec. 6).

1.3 Our Approach: Prototype-Driven Fair FL

- **Unified carrier.** We construct class-group prototypes $\mu_{g,l} = \mathbb{E}[z \mid g, y = l]$ that encode both class and group information, generalizing class-only prototypes used in prior FL work (e.g., FedProto). Prototypes are feature means: they summarize distributional information without exposing individual samples or group statistics.

- **Mechanism 1: privacy-friendly perception (←C1).** Class-group prototypes replace explicit group statistics as the communication content, and L_{PA} contracts Δ_{rep} in the feature space, controlling the EO *upper* bound (Thm. 6, Thm. 7) without perturbing the classifier. *Boundary statement:* local training still requires sensitive-attribute labels; our privacy claim is at the communication level, with a local-DP extension in App. F (optional).

- **Mechanism 2: fairness–accuracy co-optimization (←C2).** L_{conf} minimizes the prototype-level confidence difference, controlling the EO *lower* bound (Thm. 10); L_{l2l} explicitly optimizes classification of local prototypes, preserving accuracy (Lem. 13).

- **Mechanism 3: local-to-global propagation (←C3).** Bidirectional prototype–classifier alignment ($L_{l2g} + L_{g2l}$) anchors both encoder and classifier to globally consistent coordinates, and server-side post-training calibrates the aggregated classifier with global prototypes (Thm. 14).

1.4 Contributions

1. **Conjectures with theory.** We formulate three falsifiable conjectures (C1–C3) that organize the perception–adjustment– transmission space of fair FL, and support each with rigorous analysis: $\Delta_{\text{rep}} \to$ EO upper bound (Thm. 6), the contraction and residual of prototype alignment (Thm. 7), and $\delta_{\text{conf}} \to$ EO lower bound (Thm. 10); proofs in the appendix.

2. **Conjecture-driven method.** PDFFed realizes C1–C3 with three mechanisms on a unified class-group prototype carrier, each with a per-component theoretical guarantee (Lem. 13, Thm. 14).

3. **Prediction-verification empirics.** Across 3 task types, 7 datasets, 4 heterogeneity levels, and 15+ baselines, we verify predictions P1–P3 via toy experiments (quantitative theorem check), convergence and mechanism curves (Δ_{rep}, δ_{conf} vs. rounds), Pareto fronts, ablations, and sensitivity analyses.

2 Related Work

2.1 General Federated Learning

FedAvg, FedProx, SCAFFOLD, FedNova, FedRep, FedProto. These methods do not address group fairness (prototype-based methods lack group subdivision and fairness guarantees).

2.2 Fair Federated Learning

Perception-based (FedFB, FedRenyi, FairFed), adjustment-based (FedFair, mFairFL), and aggregation/model-level (LoGoFair, PraFFL, FedFACT). We position these in a *perception–adjustment–transmission* spectrum and identify the gap: none uses a group-aware prototype carrier that (i) avoids transmitting group statistics, (ii) provably controls EO from the feature space, and (iii) propagates local fairness to the global model.

2.3 Prototype-Based Learning

FedProto and successors use class prototypes for representation alignment; we generalize to class-group prototypes and add fairness guarantees.

3 Preliminaries

3.1 Setup

Federated setting with K clients, global model $\{f_\phi, h_\psi\}$, linear classifier $h_\psi(z) = w^\top z + b$ (Assumption 1). Binary classification $y \in \{0, 1\}$, sensitive group $g \in \{0, 1\}$.

Assumption 1 (Linear classifier)**.** *$h_\psi(z) = w^\top z + b$, decision rule $\hat{y}(z) = \mathbb{1}[w^\top z + b \geq 0]$. The encoder f_ϕ can be arbitrary.*

Definition 2 (Representation gap)**.** $\Delta_{\text{rep}}^{(l)} = \|\mathbb{E}[z \mid g{=}0, y{=}l] - \mathbb{E}[z \mid g{=}1, y{=}l]\|_2$.

Definition 3 (Equalized odds gap)**.** $\text{EO}^{(l)} = |P(\hat{y}{=}1 \mid g{=}0, y{=}l) - P(\hat{y}{=}1 \mid g{=}1, y{=}l)|$*;* $\text{DEO} = \text{EO}^{(1)}$*;* $\text{EO}_{\max} = \max_l \text{EO}^{(l)}$.

4 Conjectures and Theoretical Analysis

Assumption 4 (Bounded logit density)**.** *For each (g, l), $s = w^\top z + b$ has a density upper-bounded by M; after standardization by $\sigma_g = \sqrt{w^\top \Sigma_{g,l} w}$, the density is bounded by $\tilde{M}$. Gaussian features are a special case with $\tilde{M} = 1/\sqrt{2\pi}$.*

4.1 Conjecture C1: The Representation Gap Controls the EO Upper Bound

Conjecture 5 (C1)**.** *Closing the feature-space representation gap Δ_{rep} between same-label different-group samples is sufficient to close the EO gap, because Δ_{rep} controls the EO upper bound; the classifier itself need not be perturbed.*

Theorem 6 (Representation gap bounds EO)**.** *Under Assumptions A1–A4, A6 (App. A), for each label l,*

$$\text{EO}^{(l)} \;\leq\; \tilde{M}\,\|w\|\,\Delta_{\text{rep}}^{(l)} + \varepsilon_{\text{dist}},$$

where $\varepsilon_{\text{dist}}$ is a distribution-shape term that vanishes when the two groups share covariance ($\Delta_\Sigma = 0$) and shape. In the Gaussian case,

$$\text{EO}^{(l)} \;\leq\; \frac{\|w\|}{\sqrt{2\pi}\,\sigma_{\min}}\Delta_{\text{rep}}^{(l)} + \frac{\|w\|^2 \Delta_\Sigma}{2\sigma_{\min}^2}.$$

Theorem 7 (Prototype alignment contracts the gap)**.** *(Idealized) If the L_{PA} gradient acted directly on μ, then $\Delta_{\text{rep}}^{(l,t+1)} = (1 - 2\eta)\Delta_{\text{rep}}^{(l,t)}$, strictly contracting for $\eta < 1/2$. (Realistic) Under joint optimization with bounded task-gradient group difference $\|g_{\text{task},g0} - g_{\text{task},g1}\| \leq G_{\text{task}}$,*

$$\Delta_{\text{rep}}^{(l,t+1)} \;\leq\; (1 - 2\eta\lambda_{\text{PA}})\,\Delta_{\text{rep}}^{(l,t)} + \eta\,G_{\text{task}}, \qquad \limsup_{t\to\infty} \Delta_{\text{rep}}^{(l,t)} \;\leq\; \frac{\eta G_{\text{task}}}{2\lambda_{\text{PA}}}.$$

Prediction 8 (P1)**.** *Under C1 and Thm. 6–7: (a) Δ_{rep} decreases with training and stabilizes at a positive residual of order $\eta G_{\text{task}}/(2\lambda_{\text{PA}})$; (b) EO follows Δ_{rep} downward across rounds; (c) in the Gaussian toy setting, EO is quantitatively bounded by Thm. 6.*

4.2 Conjecture C2: Classifier Confidence Difference Forces the EO Lower Bound

Conjecture 9 (C2)**.** *If the shared classifier systematically assigns high confidence to one group's prototypes and low confidence to the other's, then EO is forced to be large; the prototype-level confidence difference* $\delta_{\mathrm{conf}} = \sum_l |\sigma(|\mu'_{g0,l}|) - \sigma(|\mu'_{g1,l}|)|$ *is a faithful and statistically stable surrogate for this mechanism.*

Theorem 10 (Prototype-level confidence difference bounds EO from below)**.** *Under the low/high-confidence scenario of App. D, defining* $\delta^{(l)}_{\mathrm{conf}} = |\sigma(|\mu'_{g0,l}|) - \sigma(|\mu'_{g1,l}|)|$*,*

$$\mathrm{EO}^{(l)} \gtrsim C \cdot \delta^{(l)}_{\mathrm{conf}} - O(\sigma_z, \Delta_\Sigma), \quad C = \Omega(1/\sigma_z).$$

Prediction 11 (P2)**.** *Under C2 and Thm. 10: (a) minimizing* L_{conf} *lowers* δ_{conf} *and, with it, DEO and* $\mathrm{EO}_{\max}$*; (b) the low/high-confidence scenario is observable in confidence distributions of group prototypes; (c) prototype-level* δ_{conf} *tracks the sample-level confidence difference up to* $O(\sigma_z)$*.*

4.3 Conjecture C3: Propagation Requires Globally Consistent Coordinates

Conjecture 12 (C3)**.** *Local fairness and accuracy improvements propagate to the global model only if encoder and classifier are anchored to globally consistent prototype coordinates; heterogeneity degrades global fairness through the mean prototype deviation* $\bar{\Gamma}$*, in a controllable (near-linear) way.*

Lemma 13 (Bidirectional alignment preserves global accuracy)**.** *Under FedAvg aggregation, if each client satisfies* $L_{g2l} \le \varepsilon$ *on all global prototypes, then the global classifier's margin on global prototypes is at least* $m(\varepsilon) = \log \frac{1-\sqrt{2\varepsilon}}{\sqrt{2\varepsilon}}$*, and the global misclassification probability decays as* $\exp(-m(\varepsilon)^2/(2(w^G)^\top \Sigma^G_{g,l} w^G))$*.*

Theorem 14 (Heterogeneity enters the global fairness bound)**.**

$$\mathrm{EO}^{(l)}_{\mathrm{global}} \le \tilde{M} \|w^G\| \left(\Delta^{G,(l)}_{\mathrm{rep}} + \bar{\Gamma} \right) + \varepsilon_{\mathrm{dist}},$$

where $\bar{\Gamma} = \sum_k p_k \sum_{g,l} \|\mu^{(k)}_{g,l} - \mu^G_{g,l}\|$ *quantifies heterogeneity (Thm.* **??** *in App. E).*

Prediction 15 (P3)**.** *Under C3, Lem. 13, and Thm. 14: (a) under high heterogeneity (*$\alpha = 0.01$*), PDFFed keeps global accuracy while suppressing global EO, unlike baselines; (b) global EO degradation grows near-linearly with* $\bar{\Gamma}$ *as heterogeneity increases; (c) server-side post-training strictly reduces global EO without hurting accuracy.*

5 Method: PDFFed

5.1 Design from Conjectures

Each mechanism of PDFFed is the concrete realization of a conjecture:

- ←C1: prototype communication (privacy-friendly perception) + L_{PA} (feature-space alignment).
- ←C2: L_{conf} (prototype-level confidence calibration), the only component that perturbs the classifier.
- ←C3: bidirectional alignment $L_{l2g} + L_{g2l}$ and server-side post-training (coordinate anchoring + global calibration); L_{l2l} preserves accuracy along the way.

5.2 Class-Group Prototypes

Per-client class-group prototypes $\mu^{(k)}_{g,l} = \mathbb{E}[z \mid g, y{=}l, \text{client}{=}k]$ are computed locally as feature means; the global prototype is the data-weighted sum $\mu^G_{g,l} = \sum_k p_k \mu^{(k)}_{g,l}$. Communication content is the prototypes (feature means), not group statistics.

Table 1: Accuracy and fairness comparison (mean±std over 5 seeds). [TODO: fill after runs]

Task	Dataset	Algorithm	ACC↑	DEO↓	$\mathrm{EO}_{\max}$↓
IMG	CelebA	FedAvg	–	–	–
		PDFFed	–	–	–

5.3 Local Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{l2l}} L_{l2l} + \lambda_{\text{l2g}} L_{l2g} + \lambda_{\text{g2l}} L_{g2l} + \lambda_{\text{conf}} L_{\text{conf}} + \lambda_{\text{PA}} L_{\text{PA}}, \tag{1}$$

where

$$L_{\text{PA}} = \tfrac{1}{2} \sum_l \|\mu_{g0,l} - \mu_{g1,l}\|_2^2, \tag{2}$$

$$L_{\text{conf}} = \sum_l \left|\sigma(|\mu'_{g0,l}|) - \sigma(|\mu'_{g1,l}|)\right|, \quad \mu'_{g,l} = w^\top \mu_{g,l} + b, \tag{3}$$

$$L_{l2l} = \mathrm{CE}(h_\psi(\mu^{\text{loc}}_{g,l}),\ l), \tag{4}$$

$$L_{l2g} = \mathrm{CE}(h_{\psi^G}(\mu^{\text{loc}}_{g,l}),\ l), \qquad L_{g2l} = \mathrm{CE}(h_\psi(\mu^G_{g,l}),\ l). \tag{5}$$

5.4 Server-Side Post-Training

After aggregation, the server calibrates the classifier on the four global prototypes with objective $L_{\text{post}} = L_{\text{cls}} + \lambda_{\text{eo}} \cdot \mathrm{EO}_{\text{proto}}$, where $\mathrm{EO}_{\text{proto}} = \sum_l |\sigma(\mu'_{g0,l}) - \sigma(\mu'_{g1,l})|$, using an adaptive step schedule (see App. G).

6 Experiments

6.1 Setup

- **Tasks/Datasets.** Image: CelebA, UTKFace, LFWA+, FairFace; Text: moji, bios; Tabular: ADULT, COMPAS, DRUG, DUTCH.
- **Heterogeneity.** Dirichlet(α): 0.01 / 0.05 / 0.1; Uniform. Clients: 20 / 30 / 40.
- **Two-tier design.** Breadth: all baselines × all datasets, low communication budget (5–10 rounds, fraction 0.1) — fixed-budget comparison. Depth: 1–2 representative datasets per task, 50–100 rounds, fraction 0.3 — convergence curves, mechanism curves, ablations.
- **Baselines.** FedAvg, FedProx, SCAFFOLD, FedNova, FedRep, FedProto, FedFB, FedFair, FairFed, FedRenyi, mFairFL, LoGoFair, PraFFL, FedFACT, FL_FairBatch, . . .
- **Metrics.** ACC, $\mathrm{EO}^{(0)}$, $\mathrm{EO}^{(1)}$, $\mathrm{EO}_{\max}$, DEO, DP (SPD), per-group ACC/TPR/FPR; 5 seeds + paired t-test / Wilcoxon.

6.2 Main Results

6.3 Verifying Prediction P1

6.4 Verifying Prediction P2

6.5 Verifying Prediction P3 and the Pareto Front

6.6 Ablations

6.7 Sensitivity Analysis

6.8 Optional: Differential Privacy

7 Conclusion

A Assumptions and Notation

B Proofs of Theorem 6 and Corollary ??

C Proofs for Fairness–Accuracy Tradeoff (Lem. 1, Thm. 3)

D Confidence Lower Bounds (Thm. 5, Thm. 10)

E Heterogeneity Bounds (Thm. ??, Thm. 14, Thm. 9)

F Privacy Extension (Thm. 10–12)

G Adaptive Calibration Steps

H Toy-Experiment Validation Plan